

Troubleshooting Guide: Gas Compressor Surge from Liquid Slugs Carried into Suction

Category: Rotating Equipment / Multiphase Flow Interaction **Unit:** Centrifugal Gas Compressor, Suction Scrubber Upstream **Tools:** Suction scrubber level/carryover trend review, compressor surge event correlation, transient multiphase flow review (upstream pipeline) **Fluid Package:** PR, used for gas-phase compressor performance calculations; upstream multiphase behavior assessed via transient pipeline flow tools as needed (see Case Study 18) **Symptom:** Compressor surging intermittently, with surge events correlating to brief suction scrubber high-level excursions rather than any compressor-side parameter

1. Quick Reference

Item	Detail
Reported symptom	Intermittent compressor surge events with no clear correlation to compressor suction/discharge pressure or load alone
Initially unclear	Whether surges originated from the compressor/anti-surge system itself or from an upstream liquid-handling issue
Actual root cause	Brief liquid slugs from the upstream pipeline periodically overwhelmed the suction scrubber's liquid handling capacity, causing momentary liquid carryover into the compressor suction, which disrupted gas density/molecular weight momentarily and triggered surge
Fix	Increased suction scrubber liquid handling margin (level control tuning and, where needed, mechanical modification); coordinated with pipeline operations on slug-prone flow regimes
Diagnostic signal	Surge event timestamps correlated precisely with brief suction scrubber high-level excursions, not with any compressor performance parameter in isolation
Prevention	Suction scrubber level trending tied to surge event log; use of transient pipeline slug prediction (as in Case Study 18) to anticipate high-liquid-carryover periods

2. Symptom

- **Compressor surging intermittently**, without a clear correlation to compressor suction pressure, discharge pressure, or load individually.

3. Why This Wasn't Assumed to Be an Anti-Surge Control Issue

Compressor surge naturally draws attention to the compressor and its anti-surge control system first (see Case Study 19 for a case where that assumption was correct). Here, though, it was worth checking a different possibility: **surge can also be triggered by a sudden, brief disruption to the suction gas itself** — such as liquid carryover momentarily changing gas density/molecular weight at the compressor inlet — rather than by the compressor's control system misjudging its own operating point on a stable gas.

4. Diagnostic Approach

Step 1 — Review compressor-side parameters at the moment of each surge event

Suction pressure, discharge pressure, and load were reviewed at each surge event and found to not, by themselves, explain the timing — surges occurred at a range of different load conditions, not a consistent operating point.

Step 2 — Review upstream suction scrubber level trend

Suction scrubber liquid level was reviewed at high time resolution around each surge event.

Finding: Each surge event correlated precisely with a **brief suction scrubber high-level excursion** — a short-duration liquid level spike, not a sustained high-level condition.

Step 3 — Investigate the source of the brief liquid level spikes

With the scrubber implicated, the upstream pipeline was reviewed for a source of intermittent liquid surges — consistent with the multiphase slugging mechanism described in the slug catcher case (Case Study 18), but here arriving in smaller volumes at a scrubber further downstream in the gas gathering system.

Step 4 — Confirm the carryover mechanism

The brief high-level excursions were confirmed to be large/fast enough to periodically **exceed the scrubber's vapor-liquid disengagement capacity for that instant**, allowing a small amount of liquid to carry over into the compressor suction line. Even a small liquid carryover event can momentarily and significantly change the effective gas density/molecular weight the compressor sees, which can be enough to push the compressor's instantaneous operating point across its surge line.

Quantitative Basis

- **9 surge events over 6 weeks**, each lasting only **15-40 seconds** — too brief to be explained by a sustained operating point shift.
- Scrubber level normally runs 30-50%; during each surge event, level spiked to **88-95% for 20-45 seconds** before receding, tracking the surge event timing to within the trend logger's resolution.
- Scrubber liquid handling was sized for 15 gpm continuous; the brief spikes were back-calculated (from level rate-of-rise) at an estimated **60-90 gpm instantaneous slug rate** — 4-6x design.
- Normal suction gas MW is 19.2. The estimated momentary liquid carryover (droplet entrainment equivalent to an ~8% liquid volume fraction) was sufficient to shift the compressor's effective inlet flow reading by an estimated **12%**, enough to cross the programmed surge control line.

5. Root Cause

Brief liquid slugs arriving from the upstream pipeline periodically exceeded the suction scrubber's liquid handling/disengagement capacity for short durations, causing momentary liquid carryover into the compressor suction. This transient change in suction gas properties was sufficient to trigger compressor surge, even though sustained suction/discharge pressure and load conditions looked normal.

6. Corrective Action

1. **Increased suction scrubber liquid handling margin** — through level control tuning improvements and, where necessary, mechanical modification to improve disengagement capacity for brief high-liquid-rate events.
2. **Coordinated with pipeline operations** regarding flow regimes/rates known to be more prone to generating these liquid slugs, informing operational awareness even where the pipeline hydraulics themselves weren't immediately changed.

7. Verification

- Scrubber mesh pad upgrade increased design liquid handling capacity from **15 to 35 gpm**; level control response was also tuned tighter.
- Surge events dropped from **9 in 6 weeks to 1 in the following 12 weeks**.
- The single remaining event correlated with a slug **roughly twice the size of any previously recorded event**, confirming the mechanism and setting the basis for a follow-up review of the upstream pipeline's slugging behavior at high flow.

8. Prevention / Long-Term Fix

- Established **suction scrubber level trending tied to the surge event log**, so any future surge can be immediately checked against scrubber behavior as a first diagnostic step.
- Where applicable, used **transient pipeline slug prediction** (the same technique as Case Study 18) to anticipate periods of higher liquid carryover risk and inform operational planning.

9. General Troubleshooting Checklist (Reusable)

- ☐ Review compressor suction pressure, discharge pressure, and load at each surge event — check whether they explain the timing on their own
- ☐ If compressor-side parameters don't explain the pattern, review upstream suction scrubber level at high time resolution around each surge event
- ☐ Look specifically for brief, short-duration level excursions, not just sustained high-level conditions
- ☐ If found, investigate the upstream source (pipeline slugging, a sudden rate change, etc.) of the brief liquid surges
- ☐ Confirm the carryover mechanism: could a brief liquid carryover event plausibly change suction gas density/molecular weight enough to cross the surge line?
- ☐ Improve scrubber liquid handling margin (level control tuning, mechanical modification) rather than only adjusting the compressor's anti-surge control
- ☐ Coordinate with upstream operations on flow regimes that generate slugging, where relevant

- ❑ Tie ongoing scrubber level trending to the surge event log so future events can be quickly checked against this mechanism

10. Key Takeaway

Not every compressor surge originates at the compressor — a brief liquid carryover event from an overwhelmed upstream scrubber can trigger surge just as effectively as a genuine compressor operating point excursion, and it won't show up if you only look at sustained suction/discharge pressure and load. Check high-time-resolution scrubber level data around each surge event before assuming the anti-surge control system itself is at fault; sometimes the compressor's control system is working perfectly and reacting correctly to a genuinely disrupted suction gas.

Related Concepts / Tags

centrifugal-compressor surge liquid-carryover suction-scrubber pipeline-slugging
anti-surge-control multiphase-flow Peng-Robinson

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.